

TripMate AI: Intelligent Group Activity & Travel Planning Platform

Powered by DeepSeek-V4 Flash & Pro Hierarchical Multi-Agent Architecture

Technical Project Proposal for Academic Supervisor

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1. Introduction, Scope & Product Overview

1.1. The Problem: Why Group Planning is Painful

Planning an outing or trip with friends—whether a weekend cafe hangout or a 4-day vacation—is notoriously frustrating in practice:

- **Too many open tabs:** Finding places requires jumping between Google Maps, food review apps, Agoda, Booking.com, and blogs for hours.
- **Clashing personal tastes:** Everyone wants something different (e.g., quiet cafe vs. lively pub, budget street food vs. fine dining). Discussions easily stall in deadlock.
- **Buried in chat:** Recommendations, links, opening hours, and polls quickly get lost under hundreds of messages in Zalo or Messenger.
- **Exhausted organizer:** A single person carries the burden of building the schedule, checking opening hours, and booking transport—and gets blamed if plans fail.

1.2. Target Audience & Operational Scope: Keeping It Practical

To keep the engineering scope clear and achievable, **TripMate AI** intentionally defines its operational boundary:

- **Target Users:** Groups of friends, university peers, young families, and colleagues planning leisure, dining, or travel together.
- **Scope (No Visa Complexity):** We focus strictly on destinations where users **already satisfy legal travel prerequisites** (e.g., domestic travel in Vietnam, regional travel within visa-free ASEAN territories, or travelers who already hold valid tourist visas).

Supervisor Rationale: As advised during evaluation, incorporating consular paperwork, embassy appointments, and international visa rules adds legal friction that distracts from software engineering. By focusing on travelers ready to explore, TripMate AI concentrates 100% on its core value: **smart itinerary creation, real-world spot discovery, booking options, and smooth group voting.**

1.3. The Solution: Two-Stage Consensus Planning Flow

TripMate AI structures group planning into a clear, two-stage consensus workflow inside a shared room with an AI guide:

1. **Capture Desires & Save to DB:** Friends chat naturally about dates, general budget, and preferred vibes. All user messages and preferences are persisted directly into PostgreSQL.
2. **Macro Plan Overview:** The AI first generates a high-level roadmap (number of days, daily focus, regional anchors).
3. **Lock Macro Plan (Macro Consensus):** The group discusses and agrees on the big picture before any specific venues are chosen.

4. **Micro-Spot Curation:** Once the macro framework is locked, the AI fills each day with real places (morning coffee, lunch, afternoon activity, dinner) clustered close together via live APIs (Google Places, Agoda, Klook).
5. **Granular Voting & Swaps:** Members vote thumbs UP/DOWN on individual stops. A downvoted spot can be swapped in one click with a pre-scored alternative.
6. **Final Plan Lock (Final Consensus):** The group confirms the finished itinerary. It is locked in the database, exportable to calendar (.ics) and PDF, and remains editable live on the go.

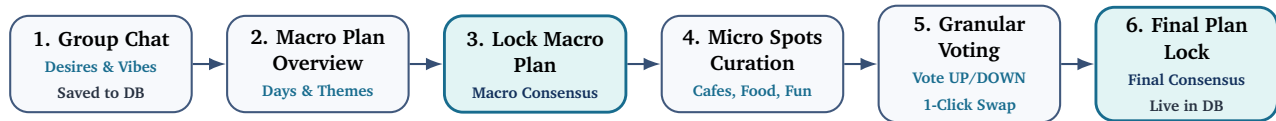


Figure 1: Two-stage hierarchical planning flow: gather desires → macro plan overview → lock macro → curate micro spots → granular vote → final plan lock

2. Platform Features & User Experience

2.1. Two Flexible Planning Scopes

- **City Hangouts & Day Outings:** Rapid curation for spontaneous weekend plans (e.g., “*afternoon specialty cafe → boardgame lounge → hotpot dinner*”). Focuses on venues open right now with minimal transit time.
- **Multi-Day Travel Vacations:** Full travel itineraries (e.g., “*4 days in Da Nang & Hoi An*”). Combines hotel suggestions and transit while ***planning every single day with the exact same detail*** (breakfast spots, morning sights, scenic cafes, dinner, and nightlife).

2.2. Shared Group Room & Saved User Data

- **Simple Link Access:** Friends join via link. Roles include *Host* (can lock the plan and manage members), *Member* (can chat, vote on stops, swap places), and *Viewer* (read-only).
- **Persistent Dialogue:** Members converse naturally; user messages, individual dietary rules (vegetarian, allergies), and budget limits are ***saved directly into PostgreSQL***, allowing the AI to maintain continuous context.
- **Event Categories:** TRAVEL, DINING, HANGOUT, ENTERTAINMENT, SIGHTSEEING, and CUSTOM.

2.3. Interactive Timeline & Live Map

- **Interactive Cards:** The schedule renders as an interactive timeline. Each card displays real photos, verified operating hours, average visit duration, price range, and direct booking links.
- **Mapbox Map Integration:** Shows clean geocoded pins, route lines, and travel times. Clicking any stop centers the map immediately.
- **Drag-and-Drop Reordering:** Members can manually drag to reorder stops or delete them, with route durations recalculated on the spot.

2.4. Two-Tier Group Consensus: Macro Lock & Micro Lock

The group always maintains democratic control through two clear stages:

- **Stage 1 (Macro Approval):** The group votes on the high-level roadmap (days, cities, themes). Once satisfied, the Host locks the macro framework.
- **Stage 2 (Micro Stop Voting):** The AI populates venues. Members vote UP/DOWN on individual stops. A downvoted spot immediately reveals 2 pre-scored alternatives with one-click swap buttons.

- **Final Consensus & Post-Lock Flexibility:** Once all legs are agreed upon, the Host hits **Confirm Plan**. Status transitions to `CONFIRMED`. Confirming does not freeze the database; the itinerary remains dynamic and can be adapted anytime during the journey.

3. The AI Agent: Communication, Workflow & Storage

3.1. Model Layer: DeepSeek-V4 Flash & DeepSeek-V4 Pro

The agent uses a two-tier model setup in Python via `**LangGraph**`:

- **DeepSeek-V4 Flash:** Fast, cost-effective model used for real-time room chat, parsing user messages, extracting travel dates/vibes, and generating quick conversational replies.
- **DeepSeek-V4 Pro:** Deep reasoning model used for multi-day schedule optimization, resolving conflicting member preferences, and clustering venues geographically to prevent zigzag travel.

3.2. Agent Architecture: User Data, System Interfacing & APIs

The AI Agent is not a standalone chatbot; it acts as the central intelligence hub connecting the user interface, backend services, database persistence, and external travel APIs:

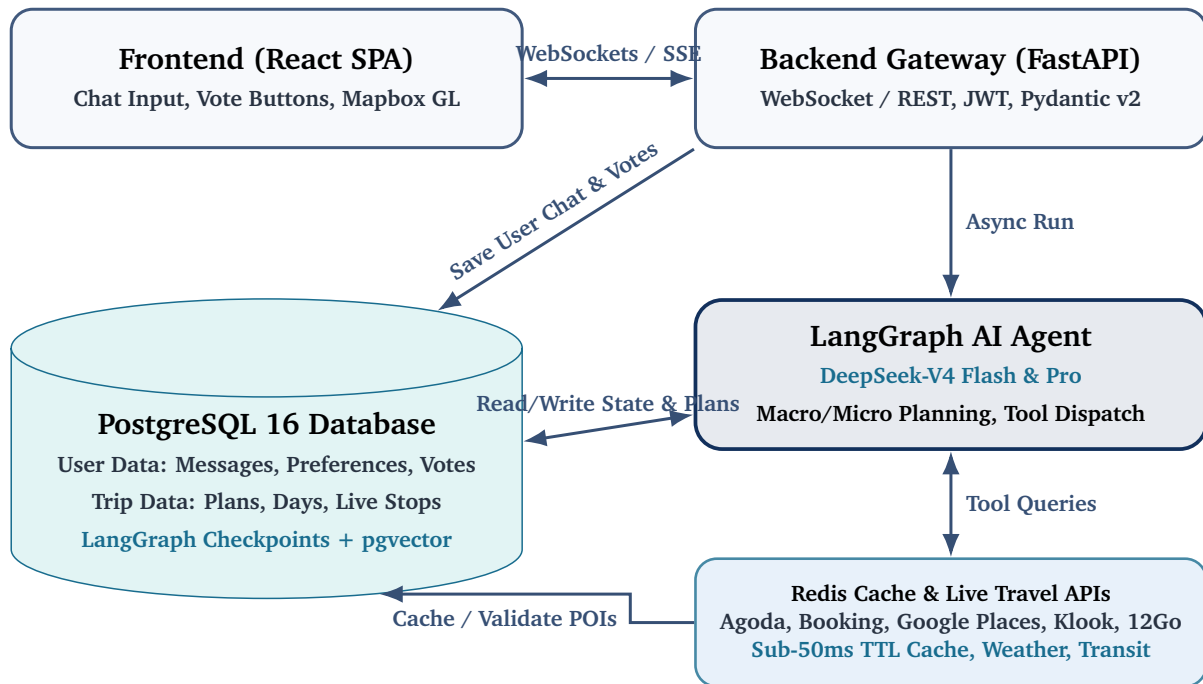


Figure 2: AI Agent architecture: user data persistence, system interfacing, and live API access

1. **Frontend Interaction (WebSockets & Streaming):** Real-time chat messages, member voting progress, and proposal cards sync instantly over WebSockets. For long answers, Server-Sent Events (SSE) stream tokens progressively to the screen.
2. **Backend Gateway (FastAPI & Pydantic v2):** FastAPI receives user requests, enforces JWT authentication and room roles, validates all input/output payloads with Pydantic v2, and dispatches tasks asynchronously to the LangGraph agent.
3. **Database Storage for User & Trip Data (PostgreSQL 16):**
 - **User Inputs Saved to DB:** User chat messages, individual preferences (budget, dietary tags), and votes are stored in PostgreSQL tables (`chat_messages`, `event_members`, `votes`).
 - **Session Checkpoints:** LangGraph's `AsyncPostgresSaver` preserves dialogue memory and reasoning checkpoints under `event_id`.

- **Relational Normalized Plans:** Itineraries are stored as clean relational records:

events → plans → plan_days → plan_stops

- **Live Atomic Updates:** Every stop is an independent database row, enabling the AI to update a single venue at any time without rewriting the entire plan.

4. External Research APIs & Curated RAG:

- **Stays (Agoda, Booking.com, Airbnb):** Live room rates, amenities, and group homestays.
- **Food & Sights (Google Places, TripAdvisor):** Verified opening hours, star ratings, and coordinates.
- **Tours & Experiences (Klook, Traveloka):** Admission tickets and guided activities.
- **Transit & Weather (Google Flights, 12Go Asia, OpenWeatherMap):** Travel schedules and forecasts.
- **Local Knowledge (pgvector RAG):** Semantic vector search over authentic local hidden gems.
- **Redis Fast Cache:** Caches frequent venue queries (2h–24h TTL) for sub-50ms speed.

3.3. Step-by-Step Workflow: From General Idea to Detailed Spots

The agent guides the group through a clear, predictable workflow:

- Step 1: Conversational Intake:** The AI asks only what is missing: destination area, dates, budget tier, and group vibe. All inputs are saved to PostgreSQL.
- Step 2: Macro Plan Overview (Big Picture First):** The AI first proposes daily themes (e.g., *Day 1: Arrival & Beach*; *Day 2: Culture & Heritage*; *Day 3: Scenic Tour*).
- Step 3: Lock Macro Plan:** The group discusses and approves this big-picture roadmap.
- Step 4: Detailed Micro Spots (Clustered & Timed):** The AI fills each day with concrete stops (morning cafe, lunch, afternoon activity, dinner), grouped within a 2–3 km radius to prevent zigzagging, with 1–2 backup alternatives per stop.
- Step 5: Granular Voting & Booking Integration:** Members vote on stops and swap places with 1-click alternatives. For complex activities (e.g., boat transfers, guided excursions, private shuttles), the AI provides direct verified booking links via Klook or Traveloka.
- Step 6: Final Plan Lock & On-the-Go Flexibility:** The Host confirms the finalized plan. The schedule remains live-editable in the database anytime (e.g., swapping to an indoor cafe if sudden rain occurs during the trip).

4. System Architecture & Operational Flow

TripMate AI adopts an asynchronous, decoupled architecture connecting the React SPA, FastAPI gateway, PostgreSQL storage, LangGraph AI orchestration, and external travel APIs.

4.1. End-to-End System Execution Flow Across All Components

The end-to-end operation coordinates all five tiers of the web system (Figure 3):

- Step 1: User Action on Frontend (React SPA):** A member types a trip idea or votes on a stop in the room. The React client applies optimistic UI rendering and sends the action payload over a secure ****WebSocket connection****.
- Step 2: Gateway Ingestion & Persistence (FastAPI):** FastAPI's WebSocket manager authenticates the JWT token, validates data schemas using ****Pydantic v2****, immediately persists messages and votes into PostgreSQL (`chat_messages`, `votes`), and launches an async AI task.
- Step 3: AI Context Loading & Planning (LangGraph):** LangGraph loads session memory from PostgreSQL via `AsyncPostgresSaver`. ****DeepSeek-V4 Flash**** generates the Macro Plan Overview (days/themes). After the group locks the macro roadmap, ****DeepSeek-V4 Pro**** curates micro spots.

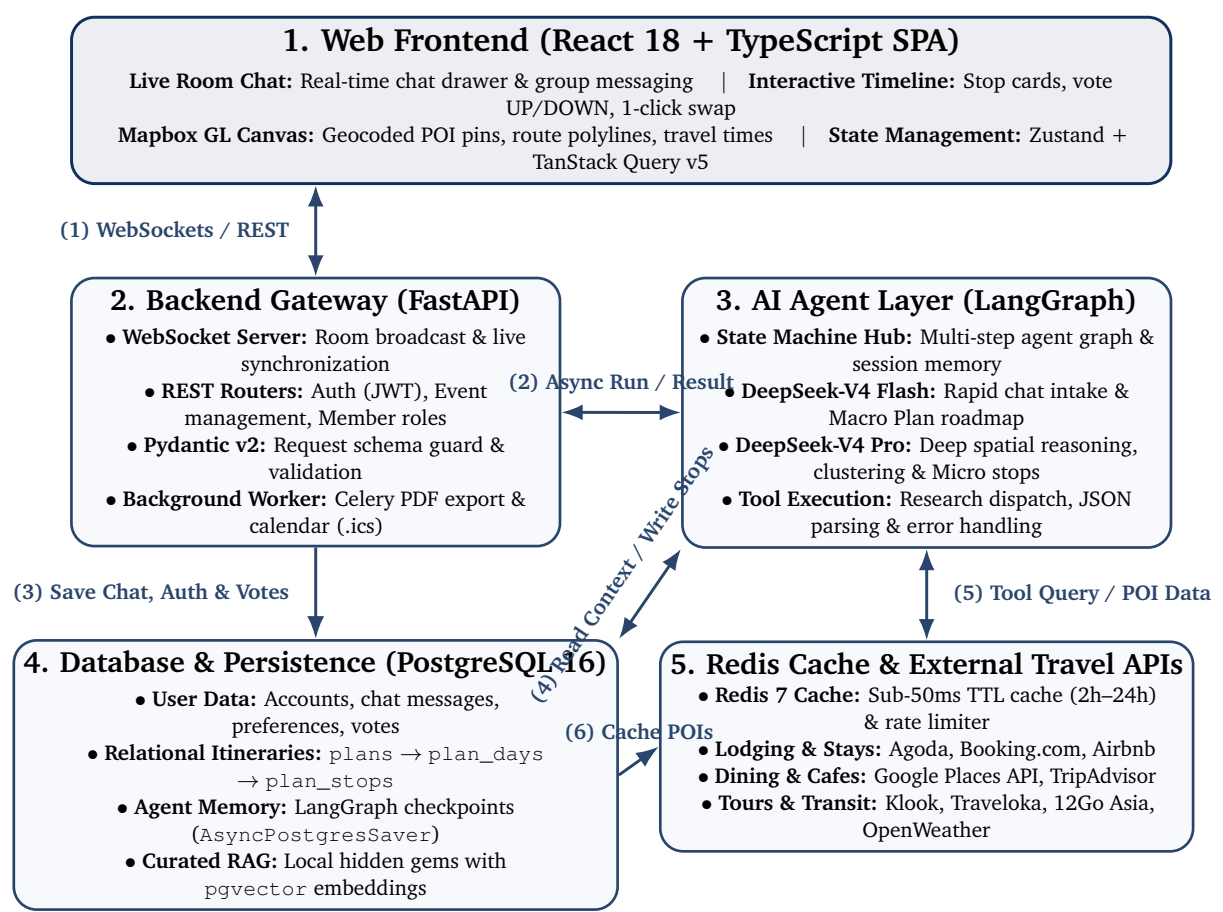


Figure 3: TripMate AI comprehensive full-stack system architecture and end-to-end data flow: spacious, readable, and structured across client, gateway, agent, storage, and external APIs

- Step 4: Real-Time Research (Redis & Live APIs):** The AI queries `**Agoda / Booking.com**` for stays, `**Google Places / TripAdvisor**` for dining/hours, `**Klook / Traveloka**` for tours, and `**pgvector**` for local hidden gems, clustering stops within 2–3 km to prevent zigzag travel.
- Step 5: Relational Storage (PostgreSQL 16):** The agent validates the plan and writes normalized records into PostgreSQL:

events → plans → plan_days → plan_stops

- Step 6: Real-Time Broadcast & Map Rendering (Backend → Frontend):** FastAPI pushes the updated itinerary event over WebSockets. The React client’s TanStack Query cache refreshes instantly, rendering the timeline cards and drawing route lines on `**Mapbox GL JS**`.
- Step 7: On-the-Go Live Modifications:** If unexpected rain occurs on Day 2 of the trip, a user simply chats: *“It’s raining, find an indoor cafe nearby”*. The agent updates only the affected `plan_stops` row in PostgreSQL, and pushes the live update to everyone’s phone screen in real time.

4.2. Technology Stack

The complete technical stack balances developer speed, type safety, and real-time collaboration:

Tier	Technology	Function & Purpose
Frontend	React 18 + TypeScript	Type-safe Single Page Application for smooth UI rendering.
	Vite	Sub-second development server and optimized production build.
	Tailwind CSS + shadcn/ui	Clean, accessible, mobile-first design tokens.
	TanStack Query v5	Server caching and instant optimistic UI feedback on votes.
	Mapbox GL JS	Interactive route mapping, custom pins, and zoom navigation.
Backend	FastAPI (Python 3.11+)	High-throughput async REST endpoints and automatic Swagger docs.
	WebSockets / SSE	Live group room chat and streaming AI token responses.
	SQLAlchemy 2.0 (Async)	Non-blocking database access with Alembic migrations.
	Pydantic v2	Fast request/response data validation and schema protection.
AI & Storage	LangGraph Python	Multi-step agent orchestration and state-machine flow.
	DeepSeek-V4 Flash	Fast chat conversations, message parsing, and tool calls.
	DeepSeek-V4 Pro	Deep spatial reasoning, clustering, and schedule optimization.
	PostgreSQL 16 + pgvector	Relational storage for events and plans; vector store for semantic RAG.
	Redis 7	Fast API cache (<50ms), session store, and rate limiter.

Table 1: Complete technology stack for TripMate AI

4.3. Deployment, Security & Rate Limiting

- **Docker Containers:** Docker Compose packages backend services, Redis, and PostgreSQL for consistent local testing and staging deployment.
- **Cloud Deployment:** Frontend hosted on Vercel edge CDN; Backend hosted on Render / Linux VPS.
- **Security:** User passwords secured with `bcrypt`; stateless JWT access tokens paired with Redis-managed refresh tokens; Room-level permissions (Host/Member/Viewer).
- **Cost Protection:** Sliding-window rate limiters in Redis prevent excessive AI token usage or runaway external API calls per room.

5. Expected Impact & Conclusion

TripMate AI solves the frustration of group planning:

- **Under 15 Minutes:** Replaces days of arguing in group chats with a fast, guided 15-minute consensus session.

- **No Burden on One Person:** Democratic voting and instant alternatives ensure everyone has a voice without one organizer doing all the exhausting work.
- **Accurate & Grounded:** Recommendations come from verified sources (Agoda, Booking, Google Places, Klook), preventing closed shops or unrealistic schedules.
- **Live & Flexible:** Because the plan lives in an active database, groups can modify their schedule anytime, before or during the trip.

TripMate AI turns group trip planning from a headache into a fun, effortless, and collaborative experience.